Time spent: 0 sec

QID: 403496

Subject: Biochemistry

System: Metabolic Reactions

Dietary carbohydrates are used to synthesize liver glycogen, which exists as particles of various sizes. During the synthesis of a new glycogen particle, a glucose molecule is linked to the protein glycogenin, after which additional glucose molecules are linked to the first glucose and to each other to form regions that are successively farther from glycogenin. Glycogen synthesis is catalyzed by the enzyme glycogen synthase, whose activity is inhibited by phosphorylation of three carboxy-terminal serine residues. During the postabsorptive state (ie, after carbohydrate absorption from a meal is finished), hormone-stimulated release of glucose residues from liver glycogen stores help maintain adequate glucose levels in the blood.

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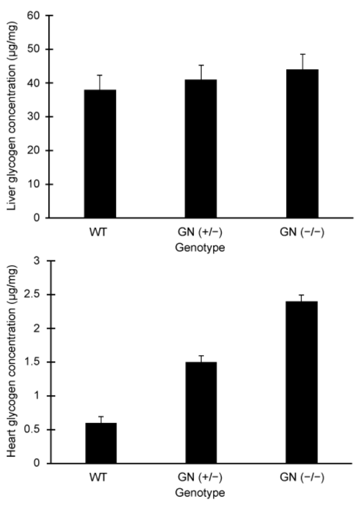


Figure 1 Liver and heart glycogen concentrations

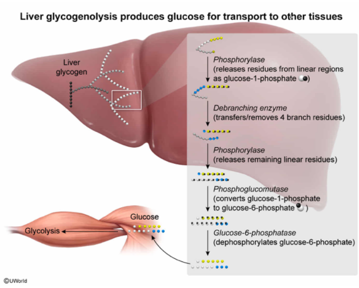
The glucagon-induced entry of glucose derived from liver glycogen into the bloodstream for transport to other tissues involves close interaction among which enzymes?

- A. Glycogen phosphorylase, debranching enzyme, glycogen synthase [10%]
- ✔ B. Glycogen phosphorylase, debranching enzyme, phosphoglucomutase, glucose 6-phosphatase [75%]
- C. Debranching enzyme, phosphoglucomutase, glycogen synthase [4%]

◯ D. Debranching enzyme, phosphoglucomutase, phosphofructokinase, glucose 6-phosphatase [9%]

Correct
75% answered correctly

Explanation:



The glucose residues stockpiled in liver glycogen particles are drawn upon to help maintain glucose levels in the blood during postabsorptive conditions. Due to the structure of glycogen (ie, consisting of both branched and linear regions), two separate enzymes are used during glycogenolysis to modify branched segments and to remove glucose residues from linear regions.

Debranching enzyme repositions branches onto the ends of short linear segments (ie, it **"linearizes" branched regions** of glycogen), releasing one glucose residue from the branch in the process. **Phosphorylase catalyzes** the **release** of **glucose residues from** the **ends** of **linear glycogen regions**.

The action of phosphorylase results in the release of glucose 1-phosphate (G1P). Within the liver, the enzyme **phosphoglucomutase** converts **G1P to glucose 6-phosphate (G6P)**. To enter the bloodstream for transport to other tissues, **G6P** must be **converted to glucose**. This conversion is accomplished by the enzyme **glucose 6-phosphatase**.

The question asks for the enzymes that closely interact to allow entry of glucose derived from liver glycogen into the bloodstream for transport other tissues. For this to occur, glycogen debranching enzyme and phosphorylase must act together to release G1P, which is subsequently converted to G6P and glucose by the actions of phosphoglucomutase and glucose 6-phosphatase, respectively.

(Choices A and C) Glycogen synthase catalyzes the addition (*not* release) of glucose residues to a growing glycogen particle.

(Choice D) Phosphofructokinase is a glycolytic enzyme and an important regulator of glycolysis, *not* the release of glucose from liver glycogen into the bloodstream via glycogenolysis.

Educational objective:
Glycogenolysis results in glucose 6-phosphate produced through the actions of debranching enzyme, phosphorylase, and phosphoglucomutase on glucose residues in glycogen. Glucose 6-phosphatase is additionally required for liver glycogenolysis to supply glucose to other tissues.

Time spent:0 sec
Subject:
Biochemistry

QID:403497
System:
Metabolic Reactions

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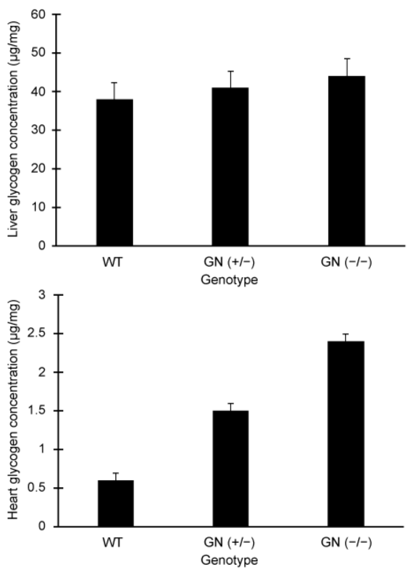


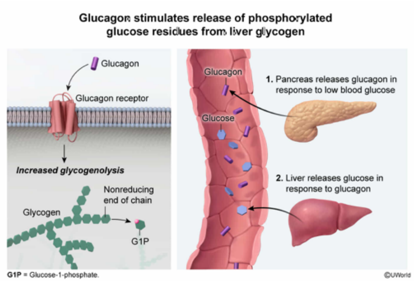
Figure 1 Liver and heart glycogen concentrations

Compared to the response that occurs during absorption of dietary carbohydrates, the response of glycogen particles to glucagon released during postabsorptive conditions is likely to result in:

- ◯ A. an increased incorporation of phosphorylated glucose molecules into glycogen particles. [8%]
- ✔ ◉ B. an increased rate of phosphorylated glucose release from glycogen particles. [48%]
- ◯ C. a decreased rate of glucose release from the nonreducing ends of glycogen branches. [8%]
- ◯ D. an increased rate of phosphorylated glucose release from the reducing ends of glycogen branches. [34%]

Correct
47% answered correctly

Explanation:



During absorption of dietary carbohydrates, glucose molecules pass through the liver, where they can be taken up into hepatocytes (ie, liver cells) and phosphorylated to produce glucose 6-phosphate (G6P). This G6P can continue into

glycolysis or it can enter other pathways, including conversion to glucose 1-phosphate (G1P) and subsequent use for glycogen synthesis.

The question asks for the response of glycogen particles to glucagon released during *postabsorptive* conditions (ie, after dietary carbohydrate absorption has ceased). Under these conditions, **liver** glycogen is broken down in the process of **glycogenolysis** to **provide glucose for** use in **other tissues**. This **process** is **stimulated by** the pancreatic hormone **glucagon**, which binds a receptor to initiate a signaling pathway leading to phosphorylated glucose release.

(Choice A) Phosphorylated glucose molecules are incorporated into glycogen particles during glycogenesis, but glucagon stimulates the reverse process (ie, release of phosphorylated glucose molecules during glycogenolysis).

(Choice C) Glucagon stimulates an *increased* rate of *phosphorylated* glucose release from the nonreducing ends of glycogen branches.

(Choice D) Glucagon stimulates an increased rate of phosphorylated glucose release from the *nonreducing* ends (*not* the reducing ends) of glycogen branches.

Educational objective:
Glucagon binding to its receptor stimulates the release of phosphorylated glucose residues from glycogen stored in liver cells.

Time spent:0 sec	**QID**:403498
Subject:	**System:**
Biochemistry	Metabolic Reactions

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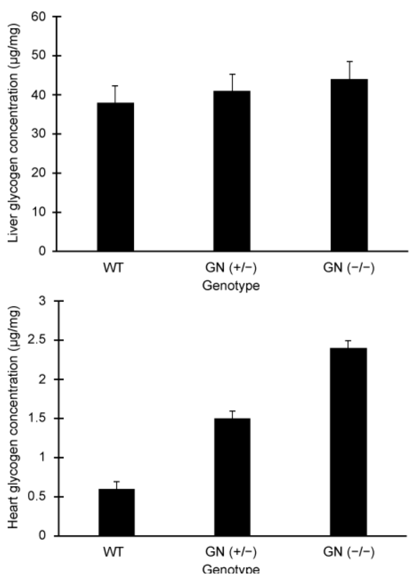


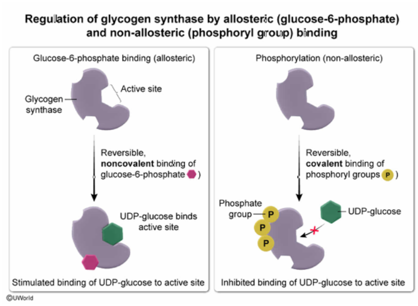
Figure 1 Liver and heart glycogen concentrations

In addition to the phosphorylation events described in the passage, glycogen synthase can be reversibly regulated by glucose 6-phosphate binding to carboxy-terminal arginine residues outside the active site. Given this fact, which type of interaction represents allosteric regulation?

- A. Phosphorylation, because phosphoryl groups bind covalently [9%]
- B. Phosphorylation, because phosphoryl group binding can be reversed by phosphatases [24%]
- C. Glucose 6-phosphate binding, because it is noncovalent [40%]
- D. Glucose 6-phosphate binding, because it occurs with basic arginine residues [25%]

Correct
40% answered correctly

Explanation:



The function of proteins (eg, enzymes like glycogen synthase) can be influenced through interactions with other molecules that bind to the protein. The binding of these modulators can occur by covalent or noncovalent interactions. **Allosteric modulators bind** their targets **reversibly** via **noncovalent** interactions at sites **outside** the **active site** (ie, allosteric sites), thereby initiating a change in the structure and function of the target protein.

The question asks which of two modulators, phosphoryl groups and glucose 6-phosphate (G6P), represent allosteric regulation. Because **G6P** is a modulator that binds glycogen synthase through reversible, noncovalent interactions outside the active site, it is an **example of allosteric regulation** of glycogen synthase activity.

(Choices A and B) Allosteric modulators bind to their targets via *noncovalent* interactions (*not* via covalent bonds). Although phosphorylation can be reversed by phosphatases, phosphorylation occurs via the formation of covalent bonds and is therefore not a type of allosteric regulation.

(Choice D) Allosteric binding sites contain residues that are complementary (ie, favorable for forming stable, reversible, noncovalent interactions) to those of the allosteric modulator binding to the site. Such binding sites are determined by their suitability for binding rather than by whether they happen to contain basic residues.

Educational objective:
Allosteric modulators influence enzyme function by binding reversibly to sites other than the active site via noncovalent interactions.

Time spent: 0 sec
Subject: Biochemistry

QID: 403499
System: Metabolic Reactions

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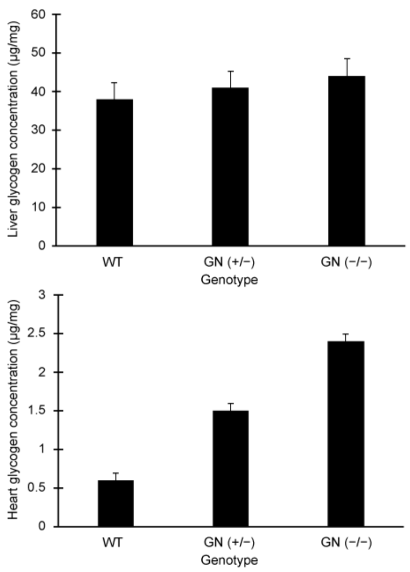


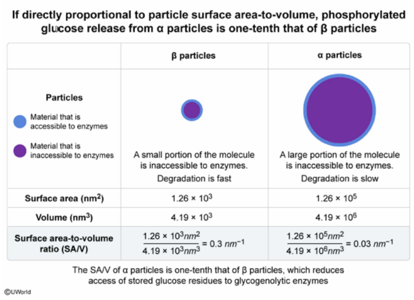
Figure 1 Liver and heart glycogen concentrations

During glycogen breakdown, if the average rate that each residue stored in liver glycogen particles is released as phosphorylated glucose is directly proportional to the particle's surface area to volume ratio, which of the following statements describes the expected relative rates of glucose 1-phosphate release from individual liver glycogen α and β particles?

- A. Release from α particles is 100 times faster than that from β particles [14%]
- B. Release from α particles is 10 times faster than that from β particles [26%]
- C. Release from α particles is the same rate as that from β particles [5%]
- ✔ D. Release from α particles is one tenth as fast as that from β particles [53%]

Correct
52% answered correctly

Explanation:



During glycogenolysis, the glucose residues in glycogen particles are released via the action of glycogenolytic enzymes. The active sites of these enzymes must physically interact with the glucose residues to catalyze these reactions. This requirement means that glucose residues buried under the surface of a glycogen particle are less accessible than those on the surface. As a result, the **fraction** of **stored glucose** molecules **accessible by glycogenolysis** is **related to** the **surface area to volume ratio of** the glycogen **particle** in which the residues are stored.

The question asks for the statement best describing the relative rates of glucose release from individual liver glycogen α and β particles, assuming the rates of release are proportional to the surface area to volume ratios of the particles. From the dimensions of glycogen α and β particles presented in Table 1, the surface area to volume ratio of α particles is one tenth that of β particles:

$$\text{Surface area to volume ratio} = \frac{\text{Surface area}}{\text{Volume}}$$

$$\alpha \text{ particle surface area to volume ratio} = \frac{1.26 \times 10^5 \text{ nm}^2}{4.19 \times 10^6 \text{ nm}^3}$$

$$\beta \text{ particle surface area to volume ratio} = \frac{1.26 \times 10^3 \text{ nm}^2}{4.19 \times 10^3 \text{ nm}^3}$$

$$\frac{\dfrac{\cancel{1.26} \times 10^5 \cancel{\text{nm}^2}}{\cancel{4.19} \times 10^6 \cancel{\text{nm}^3}}}{\dfrac{\cancel{1.26} \times \cancel{10^3} \cancel{\text{nm}^2}}{\cancel{4.19} \times \cancel{10^3} \cancel{\text{nm}^3}}} = \frac{10^5}{10^6} = \frac{1}{10}$$

Therefore, the **release from α particles** is **one tenth as fast as that from β particles** best describes the relative rates of glucose release from individual liver glycogen α and β particles.

(Choice A) The *surface area* of individual α particles is 100 times that of individual β particles, but the *surface area to volume ratio* is the value of interest specified by the question.

(Choice B) The *diameter* and the *volume to surface area ratio* of individual α particles are both 10 times that of individual β particles, but the *surface area to volume ratio* is the value of interest specified by the question.

(Choice C) Although the scientific notation *coefficients* (ie, 1.26 and 4.19) are of the same order for the surface areas and volumes of α and β particles, the *surface area to volume ratio* (and thus the phosphorylated glucose release) of α particles is one tenth (*not* the same as) that of β particles.

Educational objective:

The surface area to volume ratio of glycogen particles influences the rate at which residues can be released as phosphorylated glucose molecules.

Time spent: 0 sec	**QID**:403500
Subject: Biochemistry	**System:** Metabolic Reactions

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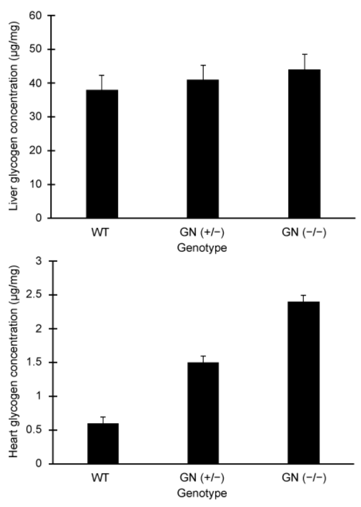


Figure 1 Liver and heart glycogen concentrations

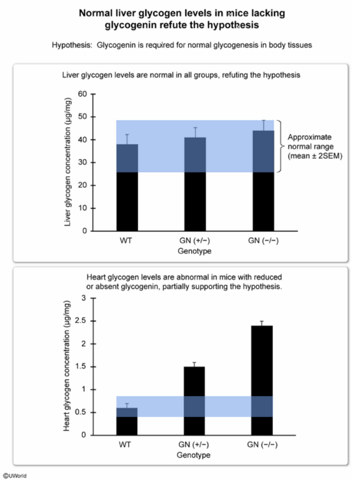
The researchers hypothesized that glycogenin is required for normal glycogenesis in all body tissues. Do the results presented in Figure 1 support this hypothesis?

- A. Yes; glycogen concentrations were normal or above normal in both liver and heart when glycogenin was present [7%]
- B. Yes; glycogen concentrations were abnormally high in hearts of heterozygous mice with reduced glycogenin expression [24%]
- ✔ C. No; glycogen concentrations were normal in the livers of all mice [57%]

D. No; glycogen concentrations were correlated with glycogenin gene expression in both liver and heart [10%]

Correct
57% answered correctly

Explanation:



Different tissues in the body have different metabolic needs and roles, and the functional importance of a protein or metabolite may vary in a tissue-specific manner. For example, in most tissues glycogen serves as a local reservoir of glucose to be used in the tissue in which it is stored. However, in the liver glycogen also represents a source of glucose that can be released into the blood for use by other tissues.

The passage describes the synthesis of new glycogen particles and presents results from a study in which glycogen levels were assessed in liver and heart samples from mice with varying amounts of glycogenin. The question asks whether the hypothesis that glycogenin is required for normal glycogenesis in all body tissues is supported by the data.

If glycogenin is required for normal glycogenesis in all body tissues, mice lacking glycogenin should have abnormal tissue glycogen concentrations. Although heart glycogen concentrations were abnormal in heart tissue from both the GN(+/−) and GN(−/−) mice, consistent with the researcher's hypothesis, **liver glycogen concentrations** were **normal in** liver tissue of **mice lacking glycogenin**. Therefore, the **hypothesis** that glycogenin is required for normal glycogenesis in *all* body tissues is **not supported by** the **results**.

(Choice A) Assessing whether glycogenin is required for normal glycogenesis requires comparing what happens when glycogenin is absent with what happens when it is present. Therefore, limiting consideration to mice in which glycogenin is present does not address the hypothesis.

(Choice B) The fact that heart glycogen concentrations were significantly higher in heterozygous mice than in WT mice suggests that glycogen concentrations are influenced by glycogenin. However, to determine whether glycogenin is *required* for normal glycogenesis in the heart and the liver requires assessing glycogen levels in both tissues in the absence of glycogenin.

(Choice D) A correlation between glycogen and glycogenin levels is not by itself enough to support the hypothesis that glycogenin is *required* for normal glycogenesis, as a correlation could be observed even if glycogenesis is normal in the absence of glycogenin.

Educational objective:
Glycogen particles are constructed by adding glucose chains successively farther from the original core, which typically contains glycogenin.

Time spent:0 sec
Subject:
Biochemistry

QID:403501
System:
Metabolic Reactions